Naitri Rajyaguru

Ph.D. Student,
Perception and Robotics Group (PRG),
Computer Vision Laboratory (CVL),
Department of Computer Science,
University of Maryland, College Park (UMD)

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Research Interests

I am interested in developing insect-inspired, minimalist cognitive architectures that enable resource-constrained robots to integrate purposive perception, memory, and action using computational imaging and computer vision.

Education

University of Maryland (UMD), College Park — Aug 2024 - Present
Ph.D. in Computer Science — GPA – 3.78/4.0
Advisor: Prof. Yiannis Aloimonos at Perception and Robotics Group

University of Maryland (UMD), College Park — May 2023
Master of Engineering in Robotics — GPA – 3.7/4.0

Gujarat Technological University, India — May 2019
Bachelor of Electronics and Communication Engineering — CGPA – 8.76/10
Gold Medalist

Awards and Fellowships

UMD CS Outstanding Achievement Award — 2025 – 2026
Future Faculty Fellowship — 2025 – 2027
UMD Dean's Fellowship — 2024 – 2026
Devang Mehta IT Award for Excellence — 2016, 2017, 2018, 2019

Publications

• **N. Rajyaguru***, A. Tajne*, S. Jardim, C. Fermuller, Y. Aloimonos. "RingNav: Task-Shaped Circular Memory for Visual Heading," *Under-Review* **[Paper Link]**, **[Website]**

• **N. Rajyaguru**, T. Wang, A. Tajne, B. He, J. Wu, C. Fermuller, C. Metzler, Y. Aloimonos. "PolarDepth: Polarization-Guided Monocular Depth for Visual Odometry," IEEE Robotics and Automation Letters (**RA-L, 2026**)**[Paper Link]**, **[Website]**

• **N. Rajyaguru***, S. Shah*, C. D. Singh, C. Metzler, Y. Aloimonos. "CodedVO: Coded Visual Odometry," IEEE Robotics and Automation Letters (**RA-L, 2024**); presented at ICRA40 2024. **[Paper Link]**, **[Website]**

Research And Work Experience

Computer Vision Research Engineer — Jul 2023 – Jan 2024
ZUPT LLC, United States
• Researched deep learning methods for high-precision object pose estimation in low-textured and challenging underwater environments.
• Designed an underwater Blender scene to conduct Sim2Real experiments for object detection and pose estimation.
• Developed a dynamic LiDAR simulator to evaluate the accuracy of different LiDARs in scene representation.

Graduate Research Assistant — Aug 2021 – May 2023
Perception and Robotics Group, UMD — **Advisor: Yiannis Aloimonos**
• Performed drone navigation experiments in unknown environments by identifying free space using the *aleatoric* uncertainty in optical flow as input. The work is published in **Science Robotics** journal.

*Equal Contributions.

• Contributed to ongoing research in 3D vision and active perception by simulating scenarios in Blender.

Perception Research Intern Jun 2022 – Aug 2022
Ford Motor Company, United States
• Conducted research and developed a pipeline for pseudo-object removal, semantic and depth inpainting, HD map generation, and enhanced localization using classical methods (Navier-Stokes, Fast Marching) and deep learning (GAN) techniques.

Research Engineer (Associate) Feb 2021 – July 2021
Swaayatt Robots, India
• Researched on Visual and LiDAR odometry pipeline for precise localization in self-driving cars.
• Independently developed LiDAR Odometry and Mapping (LOAM) pipeline, ensuring accurate sensor fusion capabilities.

Teaching Experience

• Teaching Assistant: CMSC733 Classical and Deep Learning Approaches for Geometric Computer Vision, UMD Jan 2026 – May 2026
• Teaching Assistant: CMSC426 Computer Vision, UMD Aug 2025 – Dec 2025
• Teaching Assistant: CMSC422 Introduction to Machine Learning, UMD Jan 2025 – May 2025
• Teaching Assistant: CMSC426 Computer Vision, UMD Aug 2024 – Dec 2024
• Teaching Assistant: ENEE 408U Unmanned Aerial Vehicles, UMD Jan 2023 – May 2023
• Teaching Assistant: INST 750 Advanced Data Science, UMD Jan 2023 – May 2023
• Lead Peer Research Mentor: FIRE198 Autonomous Unmanned Systems , UMD Jan 2022 – May 2022
• Robotics Tutor : Introduction to Robot Modelling , UMD Oct 2021 – Dec 2021

Mentoring/Advising

Aryan Tajne Dec 2024 to Present
Currently B.Sc. Student at the University of Maryland
Joshua Au April 2024 to Present
Currently B.Sc. Student at the University of Maryland
Alec Luterman Sep 2024 to March 2024
Currently M.S. Student at the University of Maryland

Professional Services

IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI) Journal Reviewer 2025
IEEE International Conference on Robotics and Automation (ICRA) Reviewer 2024 – Current
Nature Communications Reviewer 2025
ACM Computing Surveys Reviewer 2025
IEEE International Conference on Intelligent Robots and Systems (IROS) Reviewer 2024 – Current
TVCJ: The Visual Computer Journal (Springer) Reviewer 2025

Key Software Skills

Programming Languages: MATLAB, Python, C, C++, Embedded C
Libraries and Tools: TensorFlow, Keras, PyTorch, NumPy, Pandas, Robot Operating System (ROS), ONNX, TensorRT, OpenCV, Open3D, CUDA, Rviz, Gazebo, MoveIt, Linux
Deep Learning Architectures: VGG16, ResNet, GANs, HomographNet, SfMLearner, Attention-Based Fusion, LSTM, Position Map Regression Network, VAE, Transformers, NeRF, Diffusion Model (DDPM)
Domain Skills: Robot Perception, Localization, Deep Learning, Computer Vision, Sensor Fusion, Calibration, Mapping, Artificial Intelligence